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ai Energy of higher level – Energy of lower level = energy of photon emitted (or product of planck's constant and frequency of photon) .

aii The energy level (or orbit) of hydrogen are discrete. The energy of photon emitted is the energy difference between the two energy level which is distinct. (or a unique value). Hence, the emission line spectrum comprise of discrete lines.

7a iii 1 Electrostatic force provides the centripetal force

$$F_E = F_C$$
$$\frac{e^2}{4\pi\varepsilon_o r^2} = \frac{mv^2}{r}$$
$$\frac{me^2}{4\pi\varepsilon_o r} = m^2v^2$$
$$p = \sqrt{\frac{me^2}{4\pi\varepsilon_o r}}$$

2 P = 1.99 x 10^{-24} kg m s^{-1}, wavelength = 33 x 10^{-11} m

iv If momentum of electron in radial direction is fixed, then its uncertainty is zero. Using $\Delta\times\Delta p \geq \frac{h}{4\pi}$, $\Delta\times$ would be infinite. However the above model has a fixed radius.

bi)

$$\Delta E = E_1 - E_2$$
$$= \frac{-(13.60)(Z-1)^2}{1^2} - \frac{-(13.60)(Z-1)^2}{2^2}$$
$$= -(10.2)(Z-1)^2 eV$$
$$f = \frac{E_2 - E_1}{h}$$
$$= \frac{(10.2\times1.6\times10^{-19})(Z-1)^2}{(6.63\times10^{-34})}$$
$$= (2.46\times10^{15})(Z-1)^2$$

(b)(ii) 7.72

(ii)1. Plot point at coordinates (28, 7.72) to half smallest square

2. Points evenly scattered on both sides of line

(iii) Knowing *A* = gradient

Correct value of *A* (= 0.286 x 10^4)

-AB = vertical intercept = -0.281 x 10^4

Correct value of *B* (= 0.98)

(c) Calculate

$\sqrt{\frac{1}{\lambda_\times}}$ To get the impurity as Zinc.